

Run: cl250-alpha-seg25-10e

Log-log loss curves

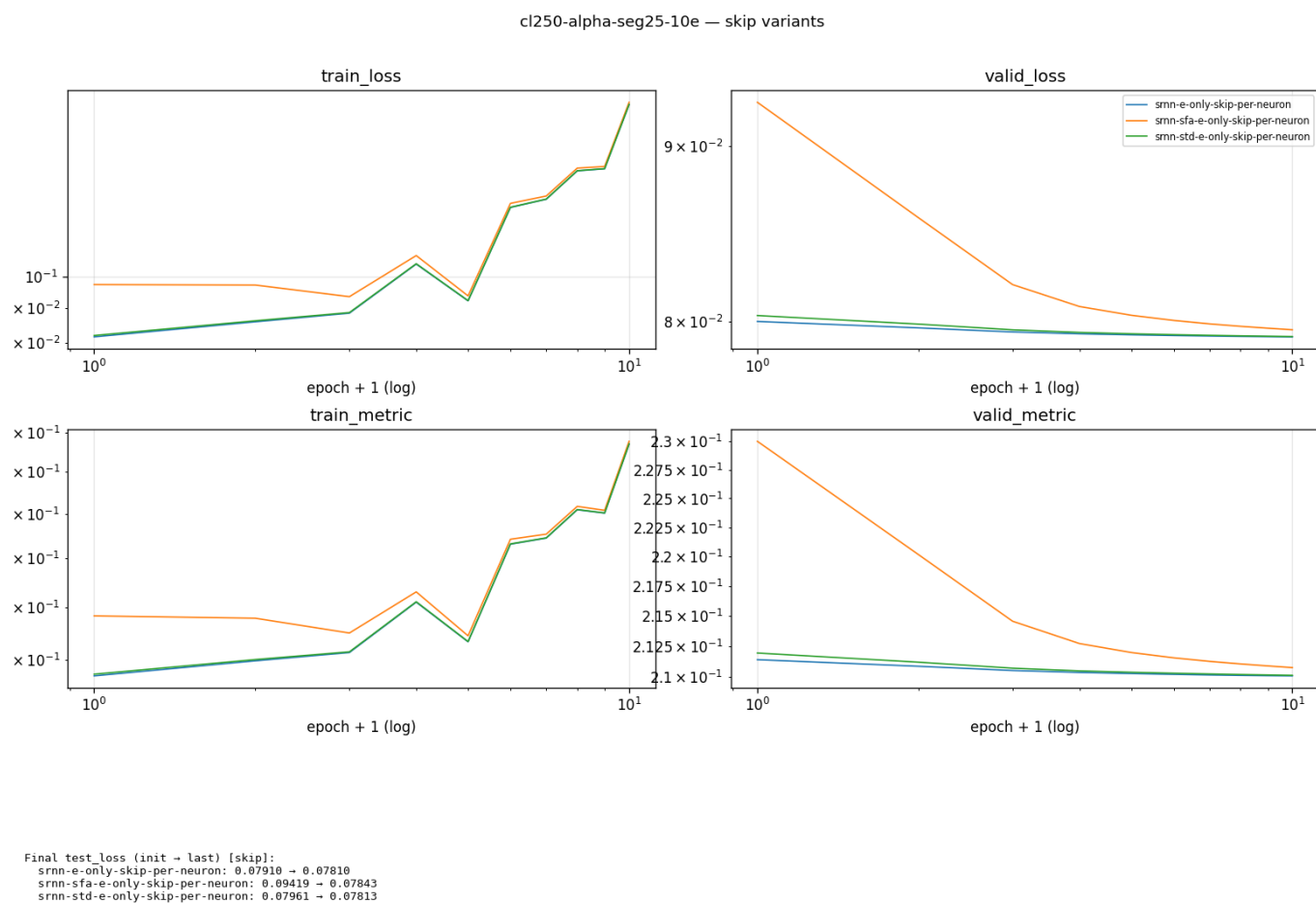
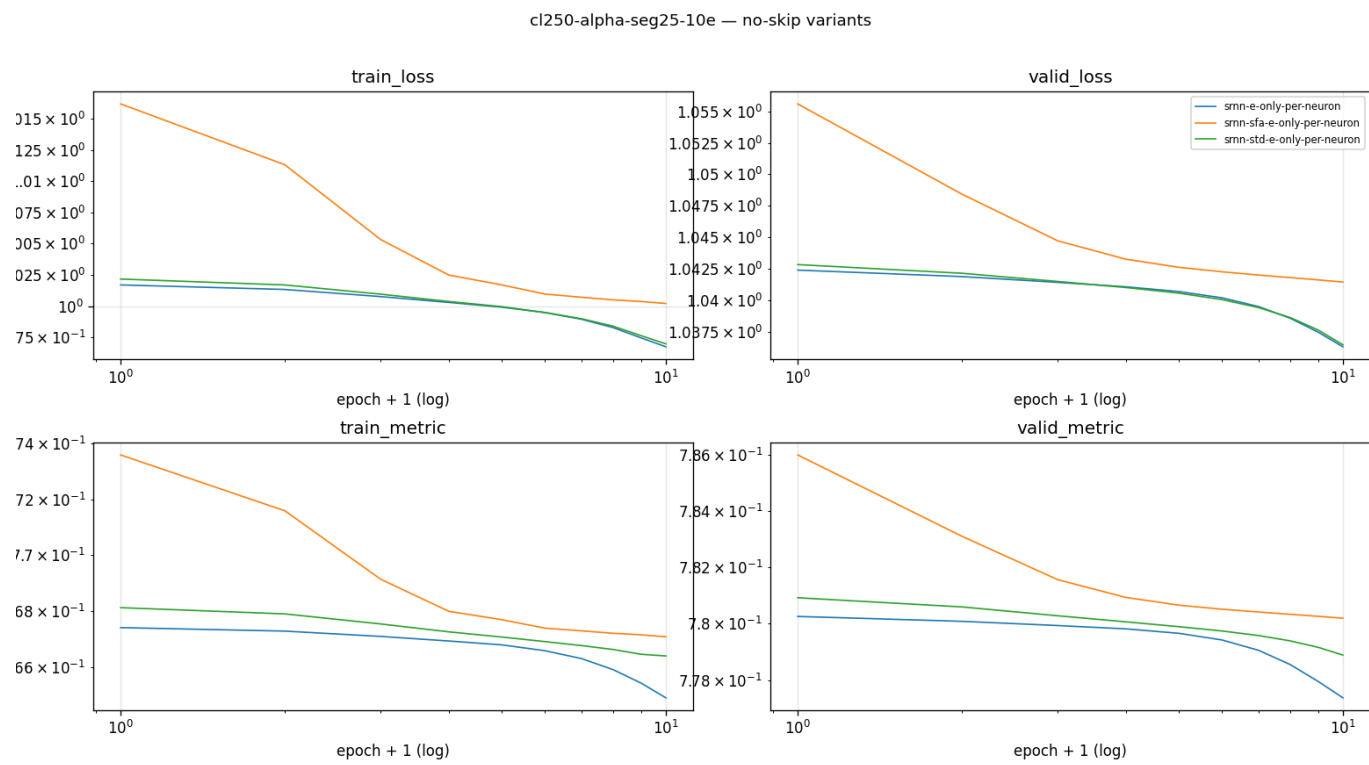


Figure 1: skip variants — log-log loss/metric



Final test loss (init - last) [no-skip]:
 srnn-e-only-per-neuron: 0.90180 - 0.89787
 srnn-sfa-e-only-per-neuron: 0.91628 - 0.90060
 srnn-std-e-only-per-neuron: 0.90234 - 0.89799

Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

cl250-alpha-seg25-10e — effective time constants (srnn-e-only-per-neuron)

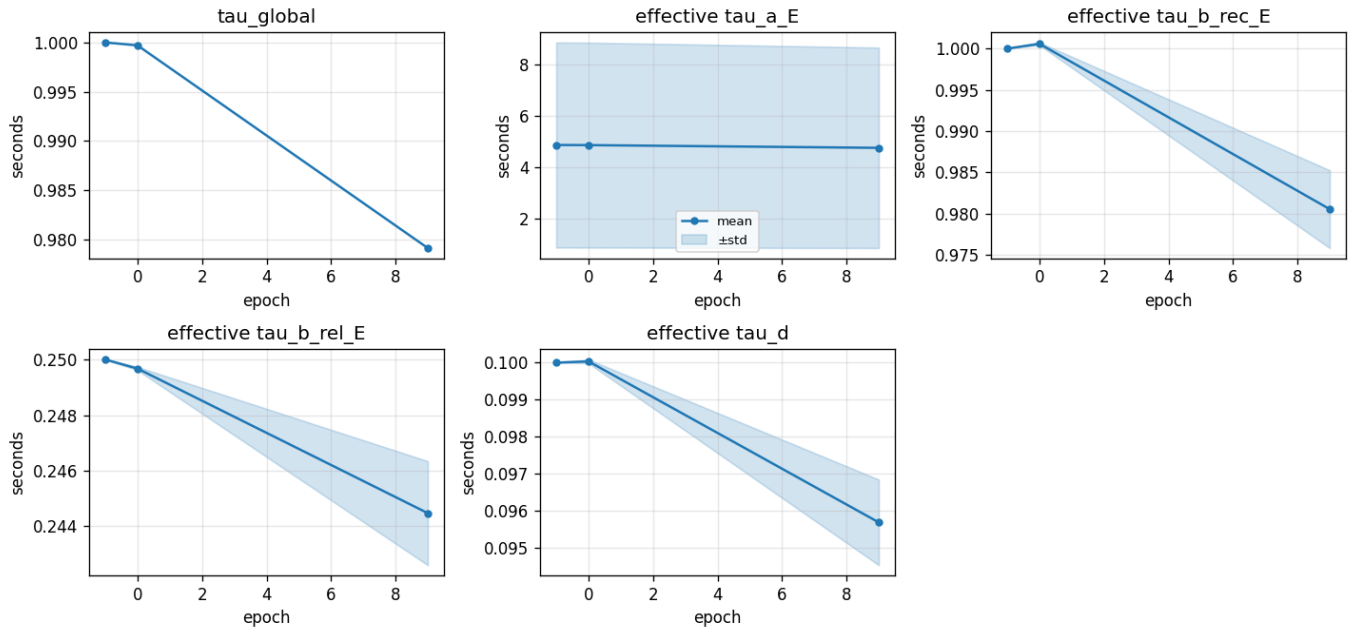
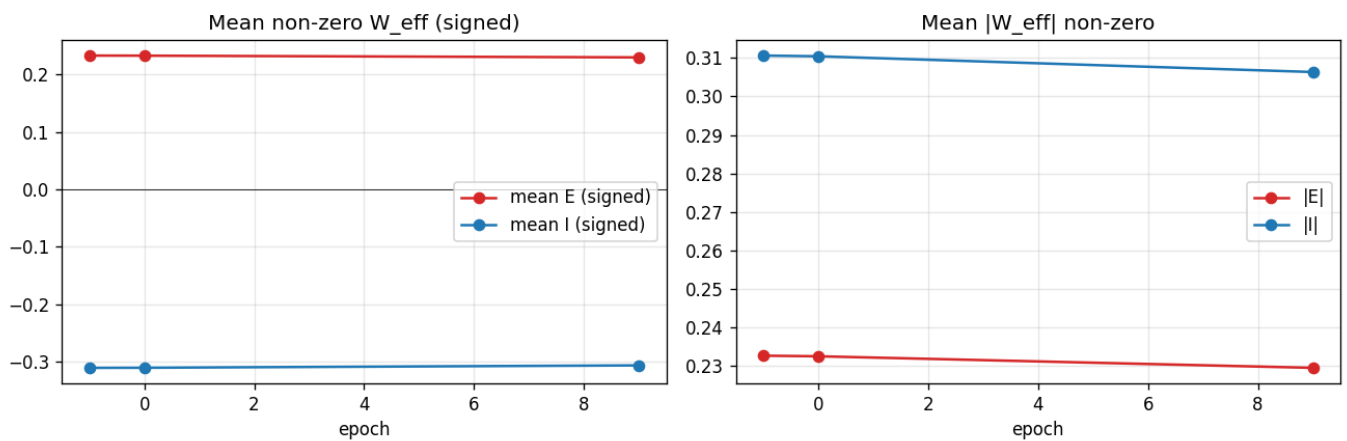


Figure 3: Tau evolution

cl250-alpha-seg25-10e — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

```
# cl250-alpha-seg25-10e - variant srnn-e-only-per-neuron (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.97915	-	-	scalar
tau_d (s)	+0.1	+0.095697	0	0.001153	300
tau_a_E (s)	+4.8777	+4.767	3.996	3.906	450
tau_b_rec_E (s)	+1	+0.98053	0	0.004719	150
tau_b_rel_E (s)	+0.25	+0.24447	0	0.001872	150
W_eff (E src, signed)	+0.23267	+0.22952	0.07641	0.07538	15048
W_eff (I src, signed)	-0.31062	-0.30635	0.07724	0.0762	14977
W_eff (E src)	+0.23267	+0.22952	0.07641	0.07538	15048
W_eff (I src)	+0.31062	+0.30635	0.07724	0.0762	14977
W_in_eff (input neurons)	-0.0003386	-0.00036664	0.1012	0.1036	6675
c_E (SFA coupling)	+0.05	+0.05011	0	0.0004296	450
c_0_E (SFA offset)	+0	-0.0020587	0	0.01032	450
a_0 (threshold)	+0.35	+0.35122	0	0.01155	300
readout_weight	-2.8854e-05	-3.1925e-05	0.06604	0.0669	6675
readout_bias	+0	+0.00081339	0	0.01411	89
ic.ic (per-variant init)	+0.098875	+0.10327	0.3842	0.3848	900

srnn-e-only-skip-per-neuron

cl250-alpha-seg25-10e — effective time constants (srnn-e-only-skip-per-neuron)

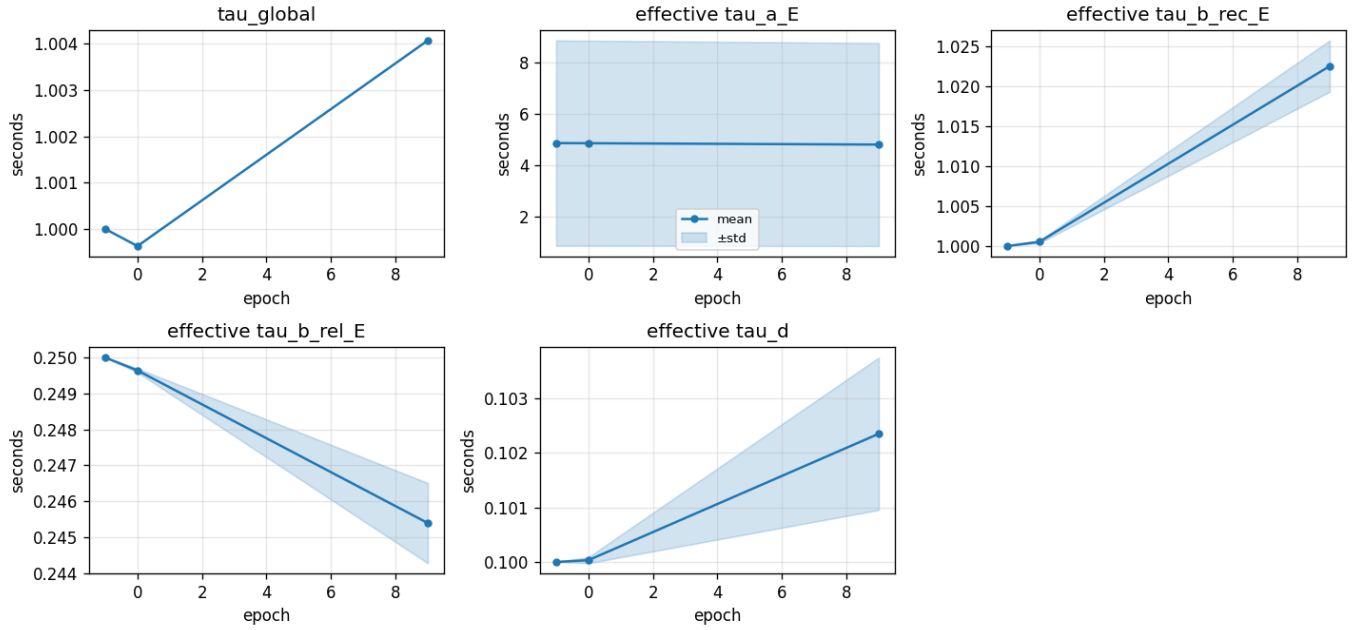


Figure 5: Tau evolution

cl250-alpha-seg25-10e — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

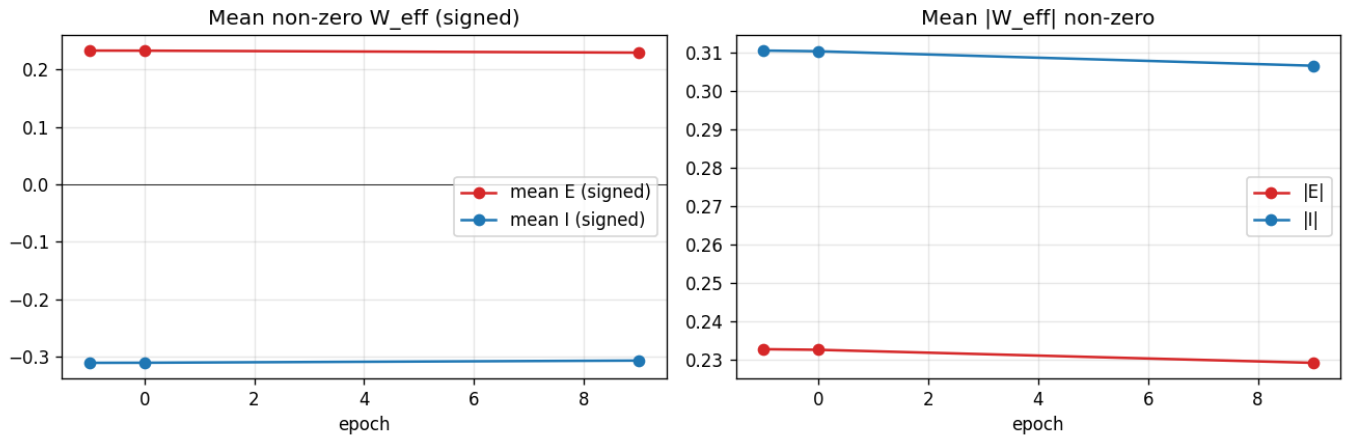


Figure 6: W_EI evolution

srnn-e-only-skip-per-neuron — parameter table

cl250-alpha-seg25-10e - variant srnn-e-only-skip-per-neuron (k=1)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+1.0041	-	-	scalar
tau_d (s)	+0.1	+0.10235	0	0.001396	300
tau_a_E (s)	+4.8777	+4.8207	3.996	3.951	450
tau_b_rec_E (s)	+1	+1.0225	0	0.003211	150
tau_b_rel_E (s)	+0.25	+0.24539	0	0.001116	150
W_eff (E src, signed)	+0.23267	+0.22911	0.07641	0.07528	15048
W_eff (I src, signed)	-0.31062	-0.30667	0.07724	0.07627	14977
W_eff (E src)	+0.23267	+0.22911	0.07641	0.07528	15048
W_eff (I src)	+0.31062	+0.30667	0.07724	0.07627	14977
W_in_eff (input neurons)	-0.00084237	-0.00093004	0.1008	0.0987	6675
c_E (SFA coupling)	+0.05	+0.051356	0	0.0003076	450
c_0_E (SFA offset)	+0	+0.028555	0	0.005884	450
a_0 (threshold)	+0.35	+0.36955	0	0.01429	300
readout_weight	+0.00078219	+0.00093101	0.06654	0.06129	6675
readout_bias	+0	+9.946e-05	0	0.002534	89
ic.ic (per-variant init)	+0.098961	+0.12409	0.387	0.396	900

srnn-sfa-e-only-per-neuron

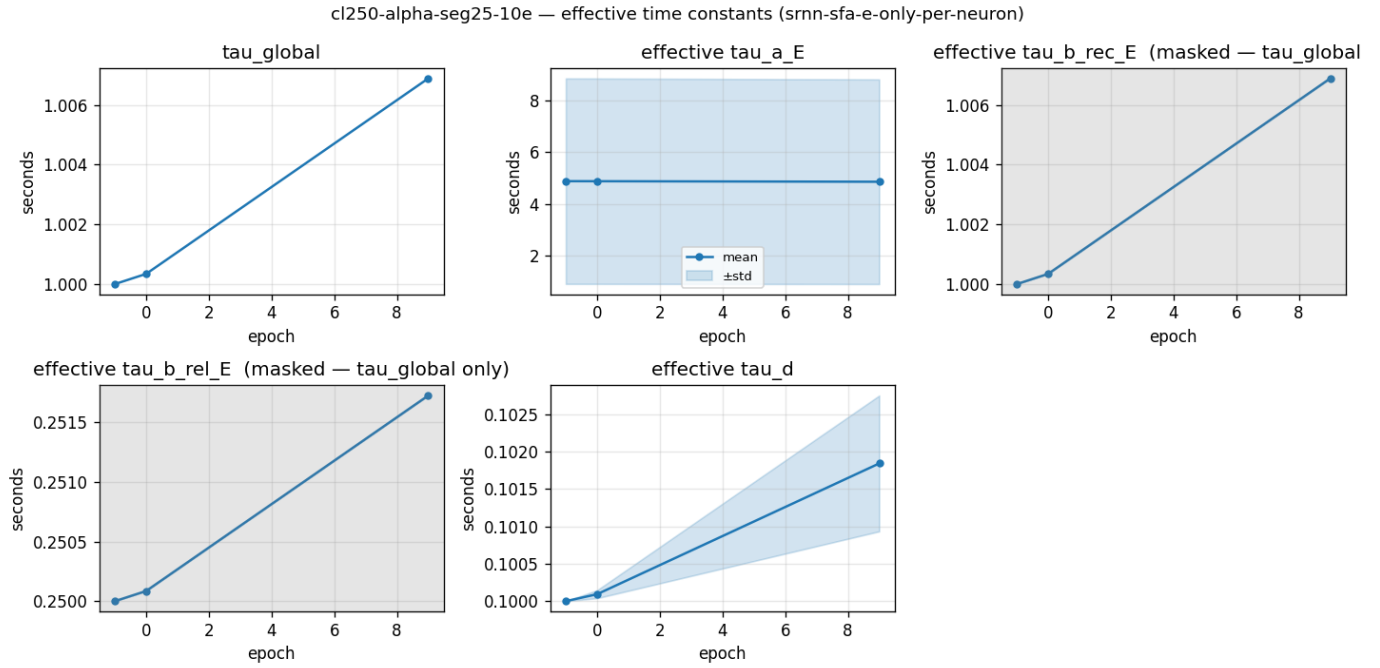
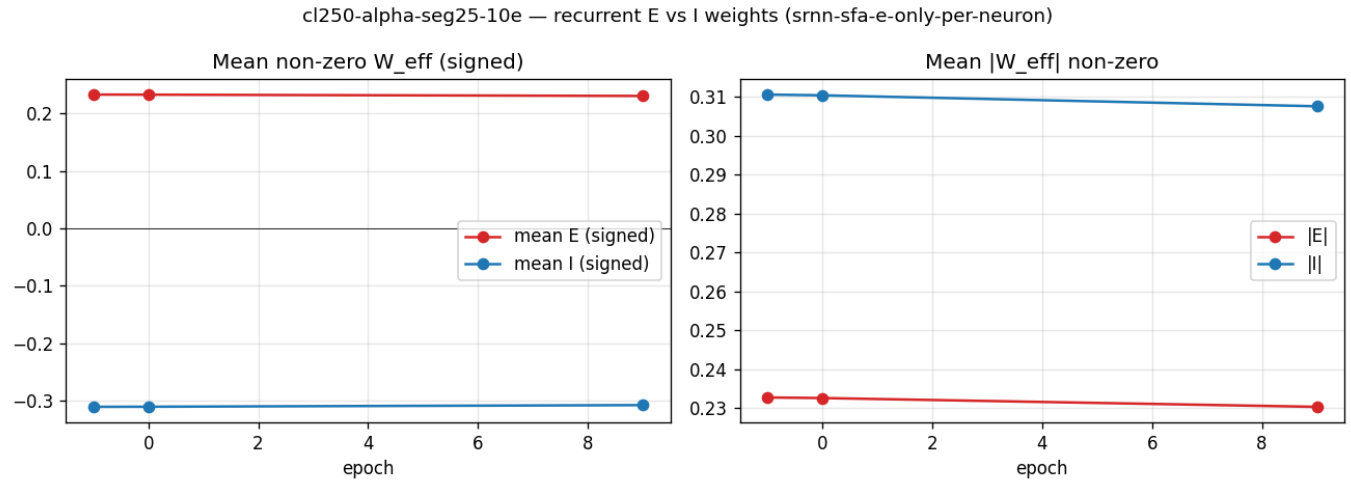


Figure 7: Tau evolution

Figure 8: W_{EI} evolution

srnn-sfa-e-only-per-neuron — parameter table

```
# cl250-alpha-seg25-10e - variant srnn-sfa-e-only-per-neuron (k=2)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+1.0069	-	-	scalar
tau_d (s)	+0.1	+0.10184	0	0.0009093	300
tau_a_E (s)	+4.8777	+4.8558	3.996	3.98	450
tau_b_rec_E (s)	+1	+1.0069	0	0	150
tau_b_rel_E (s)	+0.25	+0.25172	0	0	150
W_eff (E src, signed)	+0.23267	+0.23026	0.07641	0.07565	15048
W_eff (I src, signed)	-0.31062	-0.30761	0.07724	0.0765	14977
W_eff (E src)	+0.23267	+0.23026	0.07641	0.07565	15048
W_eff (I src)	+0.31062	+0.30761	0.07724	0.0765	14977
W_in_eff (input neurons)	-0.00015382	-6.446e-05	0.09934	0.09766	6675
c_E (SFA coupling)	+0.05	+0.050861	0	0.0002694	450
c_0_E (SFA offset)	+0	+0.019467	0	0.00589	450
a_0 (threshold)	+0.35	+0.36162	0	0.01022	300
readout_weight	-0.00098972	-0.0002994	0.06678	0.06555	6675
readout_bias	+0	+0.0010083	0	0.008808	89
ic.ic (per-variant init)	+0.13078	+0.12284	0.4172	0.4019	900

srnn-sfa-e-only-skip-per-neuron

cl250-alpha-seg25-10e — effective time constants (srnn-sfa-e-only-skip-per-neuron)

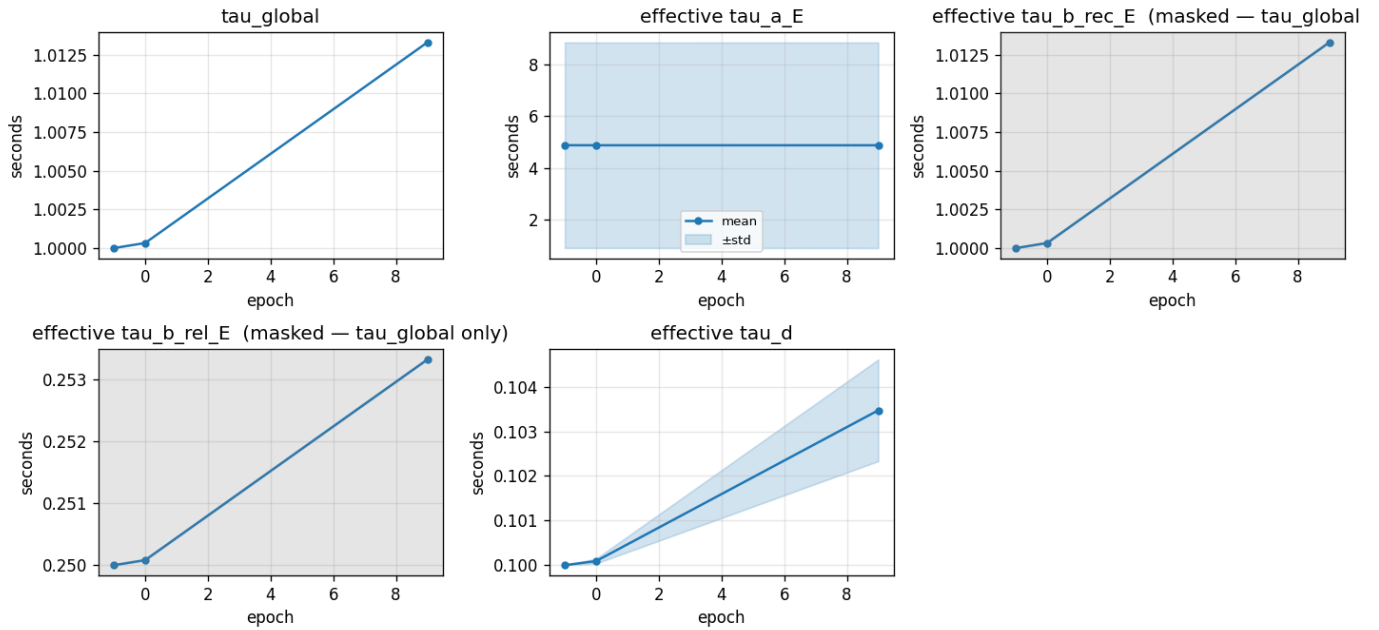
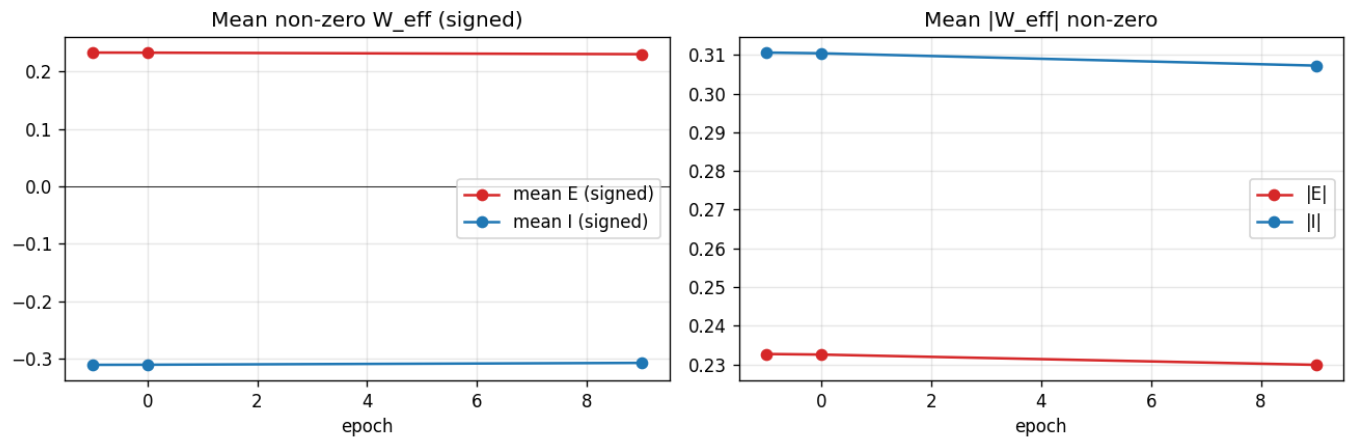


Figure 9: Tau evolution

cl250-alpha-seg25-10e — recurrent E vs I weights (srnn-sfa-e-only-skip-per-neuron)

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

```
# cl250-alpha-seg25-10e - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+1.0133	-	-	scalar
tau_d (s)	+0.1	+0.10347	0	0.001138	300
tau_a_E (s)	+4.8777	+4.8751	3.996	3.997	450
tau_b_rec_E (s)	+1	+1.0133	0	0	150
tau_b_rel_E (s)	+0.25	+0.25333	0	0	150
W_eff (E src, signed)	+0.23267	+0.22987	0.07641	0.07553	15048
W_eff (I src, signed)	-0.31062	-0.30725	0.07724	0.07641	14977
W_eff (E src)	+0.23267	+0.22987	0.07641	0.07553	15048
W_eff (I src)	+0.31062	+0.30725	0.07724	0.07641	14977
W_in_eff (input neurons)	-0.0011621	-0.00082511	0.09962	0.09705	6675
c_E (SFA coupling)	+0.05	+0.051057	0	0.0002806	450
c_0_E (SFA offset)	+0	+0.02456	0	0.005836	450
a_0 (threshold)	+0.35	+0.36477	0	0.01225	300
readout_weight	+0.00016801	+0.00010628	0.0673	0.06379	6675
readout_bias	+0	+4.3758e-05	0	0.007891	89
ic.ic (per-variant init)	+0.09231	+0.12403	0.5294	0.4013	900

srnn-std-e-only-per-neuron

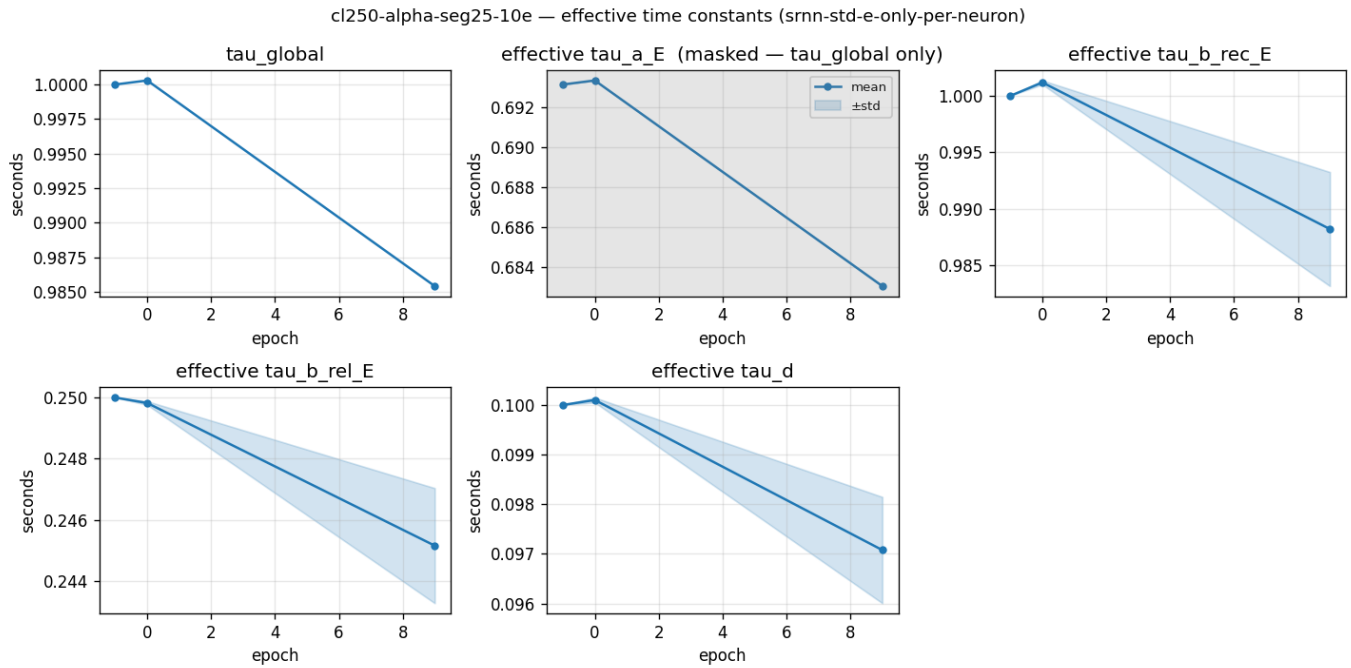


Figure 11: Tau evolution

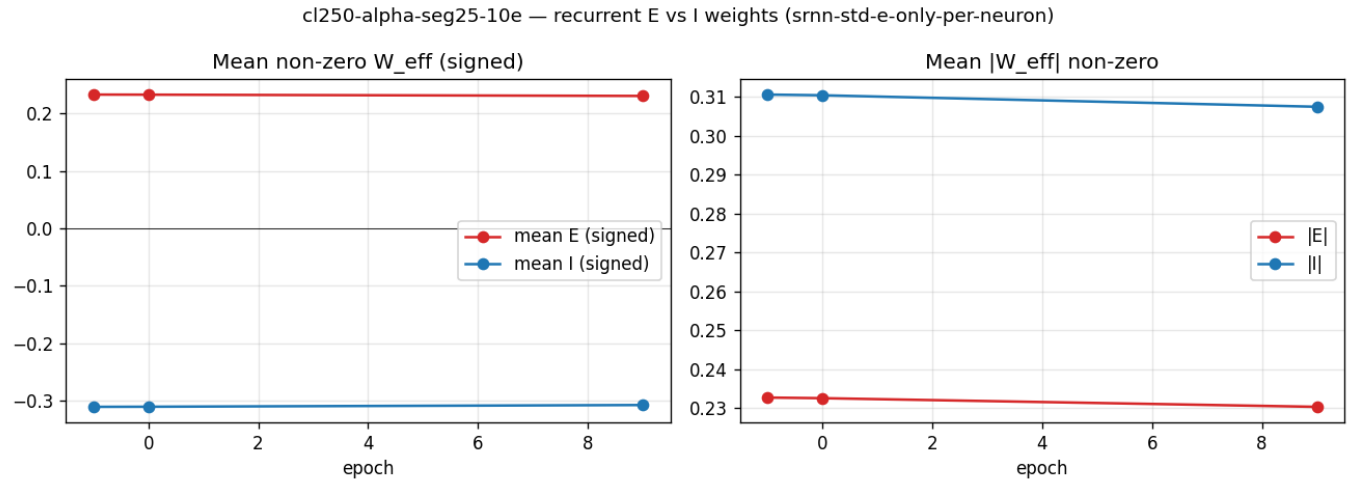


Figure 12: W_EI evolution

srnn-std-e-only-per-neuron — parameter table

cl250-alpha-seg25-10e - variant srnn-std-e-only-per-neuron (k=4)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.98542	-	-	scalar
tau_d (s)	+0.1	+0.097078	0	0.00107	300
tau_a_E (s)	+0.69315	+0.68304	0	0	450
tau_b_rec_E (s)	+1	+0.98822	0	0.005042	150
tau_b_rel_E (s)	+0.25	+0.24516	0	0.00188	150
W_eff (E src, signed)	+0.23267	+0.23029	0.07641	0.07564	15048
W_eff (I src, signed)	-0.31062	-0.30749	0.07724	0.07645	14977
W_eff (E src)	+0.23267	+0.23029	0.07641	0.07564	15048
W_eff (I src)	+0.31062	+0.30749	0.07724	0.07645	14977
W_in_eff (input neurons)	+0.00019944	+0.00032829	0.09994	0.1017	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.35382	0	0.01007	300
readout_weight	+0.00074661	+0.001915	0.06657	0.06715	6675
readout_bias	+0	-0.0024638	0	0.01364	89
ic.ic (per-variant init)	+0.087739	+0.092387	0.3868	0.3916	900

srnn-std-e-only-skip-per-neuron

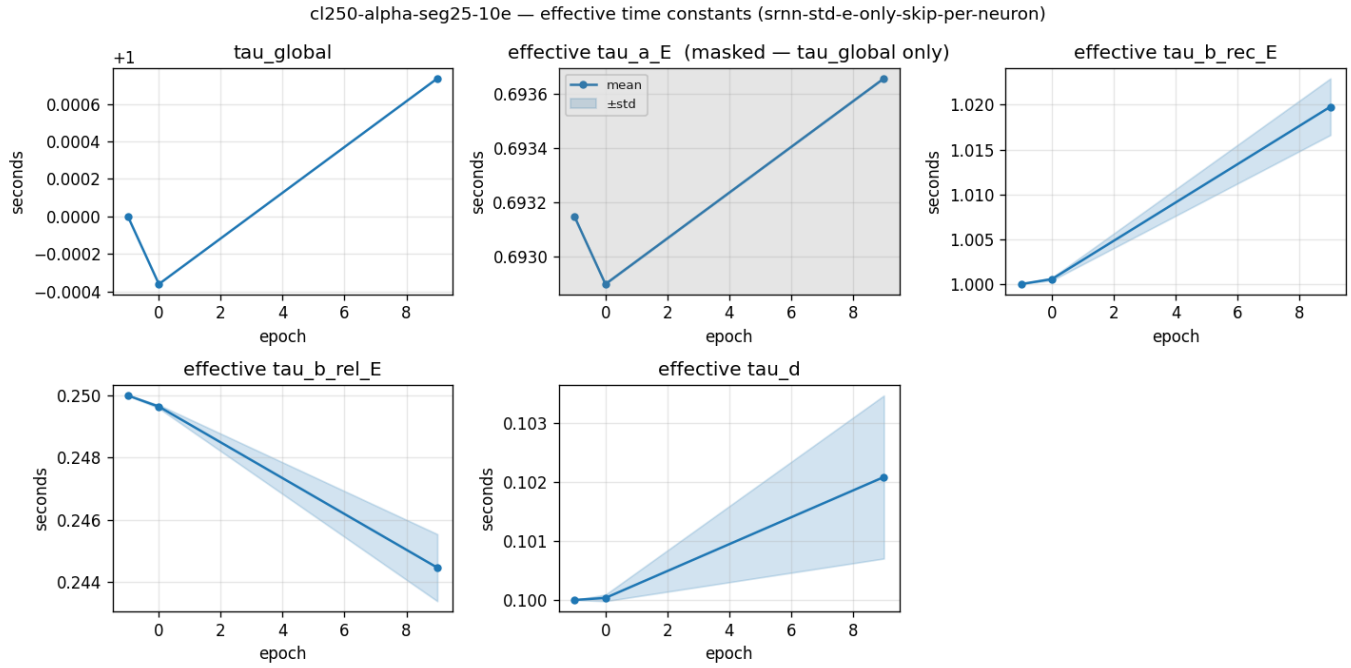


Figure 13: Tau evolution

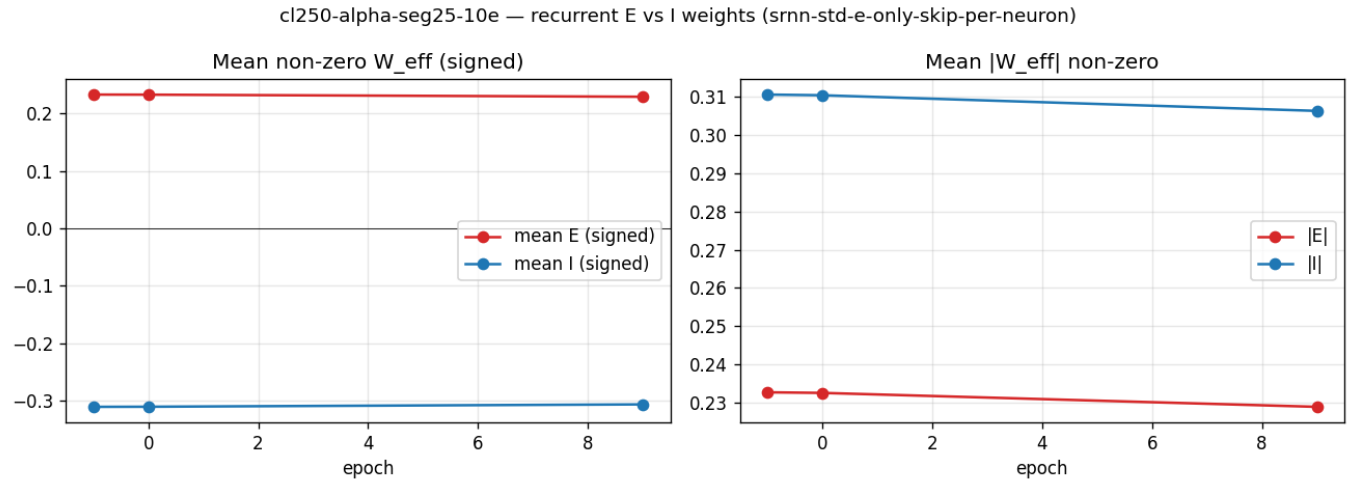


Figure 14: W_EI evolution

srnn-std-e-only-skip-per-neuron — parameter table

```
# cl250-alpha-seg25-10e - variant srnn-std-e-only-skip-per-neuron (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+1.0007	-	-	scalar
tau_d (s)	+0.1	+0.10209	0	0.001385	300
tau_a_E (s)	+0.69315	+0.69366	0	0	450
tau_b_rec_E (s)	+1	+1.0198	0	0.003154	150
tau_b_rel_E (s)	+0.25	+0.24446	0	0.001079	150
W_eff (E src, signed)	+0.23267	+0.22886	0.07641	0.0752	15048
W_eff (I src, signed)	-0.31062	-0.30635	0.07724	0.07619	14977
W_eff (E src)	+0.23267	+0.22886	0.07641	0.0752	15048
W_eff (I src)	+0.31062	+0.30635	0.07724	0.07619	14977
W_in_eff (input neurons)	+0.00036119	+0.00031513	0.09956	0.09858	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.36931	0	0.01434	300
readout_weight	+0.00055372	-0.00044298	0.06572	0.06039	6675
readout_bias	+0	-0.00058261	0	0.00371	89
ic.ic (per-variant init)	+0.089141	+0.11144	0.3857	0.3981	900